

Permutation Tests

1 Permutation Tests: Overview

Permutation tests provide a general and powerful framework for hypothesis testing when strong parametric assumptions (normality, equal variance, linearity) are hard to justify.

They are especially common in modern data settings such as:

- genomics and transcriptomics (omics data),
- neuroimaging and high-dimensional biomarkers,
- small-sample biomedical studies,
- complex outcomes where classical models may be misspecified.

The main appeal is that permutation tests are:

- **nonparametric** (no distributional form assumed),
- **exact** under the null hypothesis (for finite samples),
- **flexible** (any test statistic can be used).

1.1 Key idea: exchangeability under H_0

The core principle is *exchangeability*.

Under the null hypothesis that the two groups are generated from the same distribution,

$$H_0 : F = G,$$

the group labels “treatment” and “control” carry no information.

Thus, under H_0 :

Any observation in the pooled sample could have come equally well from either group.

This leads directly to the permutation idea:

- combine the data from both groups;
- randomly re-assign group labels while preserving sample sizes;
- recompute the test statistic under each reassignment;
- compare the observed statistic to this permutation-null distribution.

Connection to Wilcoxon. The Mann–Whitney / Wilcoxon rank-sum test is a special case of this framework: it is exactly a rank-based permutation test under continuity.

1.2 Historical note

Permutation ideas predate modern computing.

They were popularized by **R. A. Fisher** in the 1930s, especially in the context of randomized experiments.

Fisher emphasized that:

Randomization itself provides the basis for exact inference.

Permutation reasoning was also used to justify classical tests such as Student’s t test, long before computers made large-scale enumeration feasible.

Today, computing makes permutation testing routine in applied statistics.

2 Permutation Test for the Two-Sample Problem

2.1 Setup

Let

$$\mathbf{X} = (X_1, \dots, X_m) \sim F, \quad \mathbf{Y} = (Y_1, \dots, Y_n) \sim G,$$

with $\mathbf{X}$ and $\mathbf{Y}$ independent.

We want to test:

$$H_0 : F = G \quad \text{vs} \quad H_A : F \neq G.$$

Null interpretation. If H_0 is true, then all $m+n$ observations are i.i.d. from the same distribution. Therefore, the assignment of observations to the two groups is arbitrary under H_0 .

2.2 Permutation test recipe

A permutation test proceeds as follows:

1. **Choose a test statistic T .**

Examples include:

- Wilcoxon rank-sum statistic W ,
- mean difference $\bar{X} - \bar{Y}$,
- Student-type t statistic,
- any statistic sensitive to the alternative of interest.

2. **Compute the observed statistic.**

$$t_0 = T(\text{observed data}).$$

3. Generate the permutation-null distribution.

Pool the observations:

$$Z_1, \dots, Z_{m+n}.$$

Under H_0 , consider all ways of choosing m observations for group X (and the remaining n for group Y).

The number of such labelings is:

$$\binom{m+n}{m} = \binom{m+n}{n}.$$

Each labeling is equally likely under H_0 .

4. Compute permuted statistics.

Let T^* denote the statistic computed under a random reassignment of labels. Then T^* has the permutation-null distribution.

5. Compute the p-value.

For a one-sided test:

$$p\text{-value} = \mathbb{P}_{H_0}(T^* \geq t_0).$$

For a two-sided test, we use:

$$p = \mathbb{P}_{H_0}(|T^*| \geq |t_0|).$$

6. Decision.

A small p-value indicates that the observed statistic is unlikely under H_0 , providing evidence against equality of distributions.

Remark 2.1 (Exact vs Monte Carlo computation). Permutation tests can be implemented:

- **Exactly**, by enumerating all $\binom{m+n}{m}$ labelings (feasible only for small samples),
- **Approximately**, by Monte Carlo sampling of a large number of random permutations.

Monte Carlo permutation p-values become extremely accurate once the number of permutations B is moderately large (e.g., $B = 1000$ or $B = 10,000$).

Nice visual illustration of permutation test: <https://www.jwilber.me/permutationtest/>.

3 Example: Mouse Survival Data (ET Table 2.1)

3.1 Data

Sixteen mice were randomly assigned to treatment or control. Survival times (in days) after surgery were recorded.

Scientific question:

Did the treatment prolong survival?

```
ETtable21 = list(  
  treatment = c(94, 197, 16, 38, 99, 141, 23),  
  control    = c(52, 104, 146, 10, 51, 30, 40, 27, 46)  
)  
lapply(ETtable21, length)
```

Thus:

$$m = 7, \quad n = 9, \quad N = 16.$$

3.2 Wilcoxon rank-sum test (rank-based permutation test)

A natural rank-based test is Wilcoxon:

```
wilcox.test(x = ETtable21$treatment,  
            y = ETtable21$control,  
            alternative = "greater",  
            exact = TRUE)
```

Output:

$$W = 36, \quad p \approx 0.3403.$$

Conclusion. There is no strong evidence that treatment survival times are stochastically larger, at least according to the rank-based test.

3.3 Permutation test with mean-difference statistic

Now consider a permutation test using the mean difference:

$$T = \bar{X} - \bar{Y}.$$

Observed statistic.

```
t.0 = mean(ETtable21$treatment) -  
      mean(ETtable21$control)  
round(t.0, 2)
```

$$t_0 \approx 30.63.$$

This suggests a numerically large difference in means, but significance must be assessed under the permutation-null distribution.

3.3.1 Exact permutation idea

Under H_0 , every reassignment of labels is equally likely.

In principle:

- enumerate all $\binom{16}{7} = 11440$ labelings,
- compute T^* for each,
- obtain the exact p-value.

Exact enumeration is possible here, though already moderately large.

3.3.2 Monte Carlo permutation approximation

In practice, we approximate using B random permutations:

```
B = 1000
z = c(ETtable21$treatment,
      ETtable21$control)

T.star = replicate(B, {
  perm = sample(z)
  x.star = perm[1:7]
  y.star = perm[8:16]
  mean(x.star) - mean(y.star)
})

mean(T.star >= t.0)
```

This yields a Monte Carlo p-value around:

$$p_{MC} \approx 0.15,$$

close to the exact permutation p-value.

```
install.packages("perm")
library(perm)
# Perform a two-sided exact permutation test for the mean difference
exact_test_result <- permTS(ETtable21$treatment ,
                           ETtable21$control,
                           exact = TRUE,
                           alternative = "greater")

# Print the result
print(exact_test_result)
```

4 Permutation vs Bootstrap Hypothesis Tests

Permutation and bootstrap tests are both resampling-based, but they answer different questions.

4.1 Permutation testing

Permutation tests generate the null distribution by *re-labeling* observations under the assumption of exchangeability:

$$H_0 : F = G.$$

Key property (exactness):

$$\mathbb{P}_{H_0}\{p_{\text{perm}} < \alpha\} = \alpha.$$

Thus permutation tests control type I error exactly (finite-sample validity).

4.2 Bootstrap testing

Bootstrap testing resamples *with replacement* from an estimated distribution, typically a plug-in version of the null.

Bootstrap p-values are:

- approximate,
- asymptotically valid under regularity conditions,
- useful when permutation exchangeability does not hold.

Remark 4.1 (When to use which?). • Use **permutation tests** when the null implies exchangeability (e.g., randomized experiments, two-sample equality).

- Use **bootstrap tests** when there is no natural permutation structure (e.g., regression residual inference, complex estimators).

Permutation tests are exact under the null; bootstrap tests are broader but rely on approximation.

5 Exchangeability

Permutation tests rely on a fundamental symmetry property:

Under the null hypothesis, the labels attached to observations should not matter.

This symmetry is formalized through the concept of *exchangeability*.

5.1 Definition

A sufficient condition for a valid permutation test is that the observations are *exchangeable under* H_0 .

Definition 5.1 (Exchangeability). Random variables $X_1, \dots, X_n$ are called *exchangeable* if their joint distribution is invariant under any permutation of the indices.

That is, for every permutation Π of $\{1, \dots, n\}$,

$$p_{X_1, \dots, X_n}(x_1, \dots, x_n) = p_{X_{\Pi(1)}, \dots, X_{\Pi(n)}}(x_{\Pi(1)}, \dots, x_{\Pi(n)}).$$

Equivalently,

$$(X_1, \dots, X_n) \stackrel{d}{=} (X_{\Pi(1)}, \dots, X_{\Pi(n)}) \quad \text{for all } \Pi \in S(n),$$

where $S(n)$ denotes the set of all $n!$ permutations.

Interpretation. Exchangeability means that the ordering or labeling of the observations carries no probabilistic information. Only the multiset of values matters.

This is precisely the condition needed for permutation tests:

If the data are exchangeable under H_0 , then every relabeling is equally likely.

Remark 5.2 (Exchangeability vs independence). Exchangeability is strictly weaker than independence:

- If $X_1, \dots, X_n$ are i.i.d., then they are exchangeable.
- However, exchangeable variables need not be independent.

For example, sampling without replacement produces dependence, but the sample remains exchangeable because the joint distribution does not depend on order.

5.2 Examples of exchangeability

- **i.i.d. observations.**

If $X_1, \dots, X_n$ are i.i.d. from a distribution F , then the joint density factorizes symmetrically:

$$p(x_1, \dots, x_n) = \prod_{i=1}^n f(x_i),$$

so the sample is exchangeable.

- **Sampling without replacement.**

Suppose we draw balls from an urn without replacement. The outcomes are dependent, but all orderings are equally likely. Thus the sample is exchangeable even though it is not independent.

- **Compound symmetry in multivariate normal models.**

If

$$\mathbf{X} = (X_1, \dots, X_n)^T \sim \mathcal{N}(\mu, \Sigma),$$

and Σ has *compound symmetry*:

$$\text{Var}(X_i) = \sigma^2, \quad (X_i, X_j) = \rho\sigma^2 \quad (i \neq j),$$

then the joint distribution is invariant under permutations of coordinates, so $(X_1, \dots, X_n)$ is exchangeable.

- **Shift-model centering.**

If

$$X \sim F(t - \mu), \quad Y \sim F(t - \nu),$$

then after centering,

$$X' = X - \mu, \quad Y' = Y - \nu,$$

both have the same distribution $F(t)$, and the centered observations become exchangeable.

- **Coin flips with fixed totals.**

If we know that in 20 flips there are exactly 17 heads and 3 tails, and we assume exchangeability, then the positions of the tails are not special: all $\binom{20}{3}$ arrangements are equally likely.

Key message. Exchangeability is the mathematical foundation behind permutation inference.

6 Permutation Test for Discrete Data: The Lady Tasting Tea

One of the most famous examples of a permutation (randomization) test is Fisher's *Lady Tasting Tea* experiment (1935), which beautifully illustrates the logic of exact testing based on random assignment.

Historical Background

The story took place in the 1920s at Rothamsted Experimental Station in England. During afternoon tea, a lady claimed that she could tell whether milk was poured into a cup before the tea or after the tea. At the time, this may have sounded like a trivial claim, but to Fisher, it was a perfect opportunity to formalize statistical testing.

6.1 Experiment (Fisher, 1935)

A lady claims she can tell whether:

- tea was poured first, or
- milk was poured first.

Fisher proposes a controlled experiment:

- 8 cups are prepared in random order.
- 4 cups have tea poured first.
- 4 cups have milk poured first.
- The lady is asked to identify the 4 cups of one type.

Null hypothesis. The lady has no ability beyond chance guessing.

Under H_0 , her selection is simply a random choice of 4 cups out of 8.

6.2 Test statistic and null distribution

Let T be the number of cups correctly identified (successes).

Possible values:

$$T \in \{0, 1, 2, 3, 4\}.$$

Under H_0 , every selection of 4 cups is equally likely:

$$\binom{8}{4} = 70$$

possible choices.

Thus T follows a hypergeometric distribution:

$$\mathbb{P}(T = t) = \frac{\binom{4}{t} \binom{4}{4-t}}{\binom{8}{4}}, \quad t = 0, 1, 2, 3, 4.$$

6.3 Exact probabilities in R

```
t = 0:4
hypergeometry =
  (choose(4,t)*choose(4,4-t))/choose(8,4)

data.frame(t=t,
  p.t=round(hypergeometry,3))
```

This gives:

$$\mathbb{P}(T = 0) = 0.014, \quad \mathbb{P}(T = 1) = 0.229, \quad \mathbb{P}(T = 2) = 0.514, \quad \mathbb{P}(T = 3) = 0.229, \quad \mathbb{P}(T = 4) = 0.014.$$

Most of the mass lies at $T = 2$, which is what chance guessing typically produces.

6.4 Critical region at 5%

At significance level $\alpha = 0.05$, we reject H_0 only for extreme outcomes.

$$\mathbb{P}(T = 4) = 0.014 < 0.05,$$

but

$$\mathbb{P}(T \geq 3) = 0.014 + 0.229 = 0.243 > 0.05.$$

Thus the 5% rejection region is:

$$T \geq 4.$$

Conclusion. Fisher would reject only if the lady identifies all 4 cups correctly.

The exact p-value for perfect classification is:

$$p = \mathbb{P}(T \geq 4) = 0.014.$$

So the test is **very conservative**.

6.5 Monte Carlo approximation (larger problems)

When the sample size is large, exact enumeration becomes infeasible. Then we approximate the permutation distribution by simulation.

```
observed =  
  c("milk", "milk", "milk", "milk",  
    "tea", "tea", "tea", "tea")  
  
t.0 = 4    # observed perfect success  
  
nperm = 10000  
matches = replicate(nperm, {  
  perm = sample(8)  
  sum(observed[perm][1:4] == "milk")  
})  
  
mean(matches >= t.0)
```

This Monte Carlo estimate converges to the exact p-value 0.014.

6.6 Increasing the number of cups

If Fisher used 16 cups (8 and 8), then:

- the number of permutations increases dramatically,
- rejection would not require perfect classification,
- Monte Carlo methods become essential.

Thus Fisher's example illustrates both the logic and scalability issues of permutation testing.

7 Fisher's Exact Test (Connection)

Fisher's exact test for 2×2 tables is another classic example of a *discrete-data permutation test*.

Key idea. Conditioning on fixed margins, all contingency tables consistent with those margins are equally likely under H_0 .

Therefore:

- the null distribution is hypergeometric,
- the resulting p-value is exact,
- no large-sample approximation is needed.

7.1 Examples in R

```
df = matrix(c(4,0,
              0,4), nrow=2)
fisher.test(df, alternative="greater")

df = matrix(c(3,1,
              1,3), nrow=2)
fisher.test(df, alternative="greater")
```

Interpretation.

- The first table corresponds to perfect separation and rejects:

$$p \approx 0.0143.$$

- The second table is much less extreme and does not reject:

$$p \approx 0.2429.$$

Takeaway. Fisher's exact test is a permutation test for discrete outcomes, just as the Wilcoxon test is a permutation test for ranks.

8 Exchangeability and de Finetti's Theorem

Exchangeability is the symmetry principle behind permutation tests: under H_0 , labels and ordering should not matter. de Finetti's theorem goes deeper by explaining what exchangeability *means* probabilistically, especially for binary data.

8.1 Binary exchangeable sequences

Consider an *infinite* sequence of binary random variables

$$X_1, X_2, \dots \in \{0, 1\},$$

assumed to be *exchangeable*. Let

$$s_n = \sum_{i=1}^n X_i$$

be the number of successes among the first n trials.

de Finetti's representation (mixture of Bernoulli models). de Finetti's theorem states that for an exchangeable binary sequence, the joint pmf can be written as a mixture of i.i.d. Bernoulli models:

$$p(x_1, \dots, x_n) = \int_0^1 \theta^{s_n} (1 - \theta)^{n-s_n} dF(\theta),$$

for some distribution function F on $[0, 1]$.

Interpretation. This is a Bayesian-looking representation:

- θ plays the role of an *unknown success probability*.
- F is a mixing distribution (often interpreted as a prior distribution on θ).
- Conditional on θ , the observations become i.i.d. Bernoulli(θ).

More explicitly, conditioning on θ yields

$$\mathbb{P}(x_1, \dots, x_n \mid \theta) = \theta^{s_n} (1 - \theta)^{n - s_n},$$

so that

$$(X_1, \dots, X_n) \mid \theta \stackrel{iid}{\sim} \text{Bernoulli}(\theta).$$

Key message. Exchangeability does not necessarily imply independence. But de Finetti shows that *binary exchangeability is equivalent to conditional independence*:

$$\text{exchangeable (0-1) sequence} \iff \text{mixture of i.i.d. Bernoulli sequences.}$$

Why does this matter for statistics? It provides a conceptual justification for modeling dependence through latent parameters:

- If repeated trials are “similar” but not independent, exchangeability may still hold.
- Then dependence can be explained by an unobserved heterogeneity parameter θ .
- This is a foundational link between symmetry arguments and Bayesian hierarchical modeling.

8.2 Generalizations mentioned

- **Hewitt and Savage (1955):** extends de Finetti’s theorem beyond binary variables. Roughly, any infinite exchangeable sequence is a mixture of i.i.d. sequences with a suitable latent mixing measure.
- **Diaconis and Freedman (1980):** addresses *finite* exchangeable sequences. In finite samples, exact representation as a mixture of i.i.d. is not always possible, but one can obtain an approximation and finite-sample analogs. This clarifies what exchangeability implies (and does not imply) when n is fixed.

Important caveat for permutation tests. Permutation testing is a finite-sample procedure. It requires exchangeability *under the null hypothesis and under the randomization mechanism* (e.g., label shuffling is valid only if labels are arbitrary under H_0).

9 Permutation Tests in Modern Data: Neuroimaging Example

Permutation tests are especially important in modern high-dimensional settings, where we perform many tests simultaneously (often tens of thousands or more). Neuroimaging is a canonical example.

9.1 ABIDE illustration

Setting. Voxel-wise testing in neuroimaging can be framed as a two-sample problem repeated many times:

- data source: ABIDE (Autism Spectrum Disorder vs typical controls),
- subset illustration: 40 participants from a single site (e.g., Stanford),
- outcome: cortical thickness (or a similar imaging-derived measurement) at each voxel,
- goal: test whether the distribution differs between groups at each voxel.

At a given voxel k , we may compute a two-sample statistic such as:

- Wilcoxon rank-sum statistic (robust),
- mean difference or t -type statistic (if shift-in-mean is plausible),
- another voxel-wise summary T_k targeting a chosen alternative.

Thus we obtain a field of test statistics:

$$T_1, T_2, \dots, T_K,$$

where K is the number of voxels tested. (For example, $T_k = \bar{x}_k - \bar{y}_k$, where $\bar{x}_k$ and $\bar{y}_k$ are the sample mean cortical thickness values at voxel k in the autism and control groups, respectively.)

Permutation test for autism brain imaging data (Seiler 2016):<http://christofseiler.github.io/stats205/Lecture11/Neuroimaging.html>

Reference: (Nichols and Holmes 2002)(<https://onlinelibrary.wiley.com/doi/full/10.1002/hbm.1058>).

9.2 Multiple testing and permutation adjustment

If we test each voxel at $\alpha = 0.05$ *without correction*, we expect many false positives simply by chance.

Permutation methods provide a principled way to adjust for multiple testing by using the dependence structure across voxels (spatial correlation) *automatically*, without requiring strong parametric assumptions.

Core idea. Under H_0 (no group effect anywhere), group labels are exchangeable. So we repeatedly permute labels, recompute the entire voxel-wise statistic map, and use these maps to calibrate a global threshold.

Single-threshold (max-statistic) approach. Let T_k be the observed test statistic at voxel k . For each permutation i :

1. permute (shuffle) the group labels across subjects (preserving group sizes),
2. recompute the voxel-wise statistics $T_k^{(i)}$ for all voxels $k = 1, \dots, K$.

9.2.1 False Discovery Rate (FDR) control

While FWER controls the probability of at least one false positive, in high-dimensional imaging settings this can be overly conservative. An alternative is to control the *False Discovery Rate (FDR)*, defined as

$$\text{FDR} = \mathbb{E} \left[\frac{\#\{\text{false positives}\}}{\#\{\text{total rejections}\} \vee 1} \right].$$

In voxel-wise neuroimaging with K tests, FDR is often more powerful, as it allows a small proportion of false discoveries while controlling their expected rate.

Permutation-based voxel-wise p-values. For each voxel k , we compute a permutation p-value:

$$p_k = \frac{\#\{i : T_k^{(i)} \geq T_k\} + 1}{N + 1},$$

where $T_k^{(i)}$ is the statistic at voxel k under permutation i , and N is the number of permutations. Under exchangeability, these p-values are valid under H_0 .

Benjamini–Hochberg (BH) procedure. Let $p_{(1)} \leq \dots \leq p_{(K)}$ denote the ordered p-values. Define

$$k^* = \max \left\{ k : p_{(k)} \leq \frac{k}{K} \alpha \right\}.$$

Then voxels corresponding to $p_{(1)}, \dots, p_{(k^*)}$ are declared significant. This controls the FDR at level α .

Why FDR is reasonable in imaging. Neuroimaging data exhibit spatial dependence across voxels. The BH procedure remains valid under certain positive dependence structures (e.g., PRDS conditions), which are commonly satisfied in spatially correlated imaging data. Permutation-based p-values further reduce reliance on parametric distributional assumptions.

9.2.2 FWER control (global threshold).

Record the maximum statistic over voxels:

$$T_{\max}^{(i)} = \max_k T_k^{(i)}.$$

The collection $\{T_{\max}^{(i)}\}_{i=1}^N$ approximates the null distribution of the *maximum* statistic under H_0 .

Define $C = \lfloor \alpha N \rfloor$. Let $T_{\max, (C+1)}$ denote the $(C + 1)$ -th largest value among $\{T_{\max}^{(i)}\}$. Then a voxel k is declared significant if

$$T_k \geq T_{\max, (C+1)}.$$

This controls the *family-wise error rate* (FWER) at approximately α :

$$\mathbb{P}(\text{at least one false positive}) \leq \alpha.$$

Adjusted p-value (voxel-wise). A permutation-adjusted p-value can be computed for each voxel k as:

$$p_k^{\text{adj}} = \frac{\#\{i : T_{\max}^{(i)} \geq T_k\}}{N}.$$

Intuitively, this asks:

How often does the null produce *any voxel* as extreme as the observed voxel k ?

Practical note. Because p_k^{adj} is based on N permutations, its resolution is about $1/N$. For example, with $N = 10,000$, the smallest attainable nonzero p-value is about 10^{-4} .

Why max-statistic works well in imaging. Note that one could have applied Bonferroni adjustments to voxel-wise permutation-based p-values for controlling FWER as well, but that does not work well in imaging applications due to the large number of voxels.

Neuroimaging data exhibit strong spatial dependence as well. The permutation max-statistic approach automatically accounts for this dependence, whereas naive corrections assuming independence (e.g., Bonferroni) can be overly conservative.

10 Summary

- A sufficient condition for permutation test is exchangeable of observations.
- If the observations are not exchangeable, then some permutations are more likely than others.
- When doing permutation tests, in order to control the probability of type I error, one must establish that the observations are exchangeable under H_0 .
- Permutation approaches can be used for adjusting p-values in multiple testing problems.
- Different test statistics emphasize different alternatives (location shift, dispersion change, or general shape differences).
- Rank-based tests are robust and broadly applicable; they can be less sensitive than mean-based tests when a pure shift-in-mean model is truly correct.
- Mean-based statistics can be more powerful when the shift-in-mean assumption is reasonable, but are more sensitive to outliers and heavy tails.
- Permutation tests provide a unified, assumption-light framework for inference by constructing the null distribution through label reassignments.
- A sufficient condition for valid permutation inference is exchangeability of the observations (or labels) under H_0 .
- If observations are not exchangeable, then some permutations are more likely than others, and naive permutation p-values may fail to control type I error.
- In multiple testing problems (omics, neuroimaging), permutation methods can adjust for multiplicity by using max-statistics (or related global summaries) to control FWER.
- FDR control is also easy to implement based on permutation-based p-values.